

M&A OPERATING SYSTEM

Product Design

AI Analytics Platform

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1. Core Platform Capabilities

This chapter defines the target logical architecture for the AI Analytics Platform. It covers thirteen pipeline components in the order a query will encounter them, using a single portfolio manager query as a running example throughout. Each section describes what a component will do, its controls contract, and its position in the pipeline — with no references to specific technology products or vendor implementations.

Platform roles — who will interact with each component and how — are defined before the component descriptions.

Platform Roles

The platform will operate across three distinct planes: an **analytical plane** (querying, exploring, and exporting governed data), a **controls plane** (defining, approving, and administering the semantic layer and its access controls), and an **infrastructure plane** (platform deployment, health, and technical configuration).

Role	Plane	Definition
Analytical End User	Analytical	Ask governed analytical questions via natural language; receive role-constrained results without knowledge of data structures or metric identifiers
Power Analyst	Analytical	Multi-dimensional exploration, governed drilldown, lineage inspection, result export
Data Modeller	Controls	Will own semantic data definitions in the SDR: logical data elements, object models, business definitions, critical data elements, and physical schema mappings. Will ensure the organisation's data assets are accurately described and structured — the foundational layer on which metric definitions are built
Metrics Modeller	Controls	Will own semantic metrics and analytics definitions in the SMR: key performance metrics, analytics operations, trend analysis constructs, and insight definitions. Must combine domain knowledge — what does this metric mean in this business context — with modelling precision: how it is calculated, from which sources, under which dimensional hierarchies, and with which access policies
Entitlements Manager	Controls	Responsible for defining and maintaining the organisation's data entitlement policies: who may perform which actions on which data elements, analytics definitions, and business process metrics. Will configure the metric access sets, dimension access sets, row scope, and column masks that RAPL enforces at query time

Role	Plane	Definition
Analytics Governance	Controls	Overall accountability for the governance, integrity, and outcomes of the analytics platform. Will own SMR registry health, approve semantic definition changes from Metrics Modellers, oversee entitlement policy governance, and be accountable for the quality, accuracy, and completeness of analytical outputs across the organisation. The final authority on what is defined, who can access it, and whether the platform is delivering the right outcomes. Must be in place before go-live — without this role the registry has no approval authority and the platform cannot serve any query
Integration Engineer	Controls	Will register execution backends, maintain connection configuration, and declare the physical mappings that the Federated Query Engine resolves at execution time. Operates through configuration interfaces only — not the query path
Platform Admin	Infrastructure	Will be responsible for platform health, deployment, infrastructure-level governance, and technical platform configuration including controls settings, feature flags, and deployment configuration. Will implement the technical policies and settings determined by Analytics Governance. Has no query interface into analytical data

Roles are not mutually exclusive. A single individual may hold multiple roles; the platform will evaluate entitlements from the combined JWT claims present at query time.

The **Data Modeller** and **Metrics Modeller** are the critical pre-conditions for everything downstream. No analytical query can be served against a metric that has not been modelled, registered, and approved. The Data Modeller will establish the foundational data definitions in the SDR — without accurate data structure definitions, metric definitions cannot be built. The Metrics Modeller will build on that foundation to define the analytical layer in the SMR — without registered, approved metric definitions, the controls pipeline, the entitlement layer, the lineage store, and the Data Visualization Language (DVL) have nothing to operate on. Analytics Governance will hold final approval authority over both layers.

Role × Feature Access

Feature	End User	Power Analyst	Data Modeller	Metrics Modeller	Entitlements Mgr	Analytics Governance	Integration Eng	Platform Admin
Natural language query	✓	✓		✓		✓		
Role-aware results	✓	✓		✓		✓		
Governed drilldown		✓		✓		✓		

Feature	End User	Power Analyst	Data Modeller	Metrics Modeller	Entitlements Mgr	Analytics Governance	Integration Eng	Platform Admin
Lineage inspector		✓	✓	✓		✓		
Result export		✓		✓		✓		
Provenance artifacts	✓	✓		✓		✓		
SMR browsing		✓	✓	✓	✓	✓		
SDR management			✓					
Metric definition authoring				✓				
Metric definition approval						✓		
Dimension & hierarchy management				✓		✓		
Entitlement policy management					✓	✓		
Backend registration							✓	
Controls configuration								✓
Audit trail					✓	✓	✓	✓
Platform infrastructure								✓

Architecture and Request Flow

The platform will expose its capability through three consumption modes: direct API access (a host-built custom analytics UI calling the MCP Capability Layer with a structured tool invocation); conversational backend access (the AI Chat Platform calling the Analytics Platform as a tool provider); and agentic access (scheduled agents, event monitors, and automated report pipelines calling the MCP Capability Layer with

machine-issued JWTs). All three modes will share a single entry point and a single controls pipeline; consumption mode will affect only the caller's interaction pattern, not the trust model applied.

Architecture Diagram

Diagram (mmdc unavailable)

```

flowchart TD
    subgraph Consumers["AI Consumers"]
        direction LR
        ChatComp["<b>Conversational AI</b>\nConversational UI · Image/Chart rendering · Tool  
call routing\nMemory · Shared conversations"]
        CustomUI["<b>Custom analytics UI</b>\napplication hosted, embedded analysis,  
presentations, dashboards, self-service apps"]
        Agents["<b>Agentic AI</b>\nIndependent Agents - event monitors · report  
pipelines\nAnomaly Detection · Event Prediction · Data Quality · Risk Tolerance · Reviews  
etc"]
    end

    subgraph analytics["Analytics Engine"]
        direction TB
        MCP["<b>API/MCP Interface</b>\nMCP server runtime · tool/resource/prompt presentation  
· JWT validation"]
        IRA["<b>Intent Resolution Agent (IRA)</b>\nRAG over SMR catalogue · LLM intent  
ranking · confirmation gate\nnatural language → resolved operation_id + params"]
        RAPL["<b>Role-Aware Projection Layer (RAPL)</b>\nentitlement decisions · metric/  
dimension access · row scope · column masks\nreads role definitions from DES"]
        SVL["<b>Semantic Validation Layer (SVL)</b>\nSMR resolution · schema validation ·  
entitlement enforcement · LQP generation\nentirely deterministic – no AI"]
        SCL["<b>Semantic Controls Layer (SCL)</b>\ndata scale · complexity · classification ·  
compliance · concurrency"]
        PQP["<b>Physical Query Planner (PQP)</b>\nphysical_mapping resolution · execution  
plan compilation · entitlement filters applied"]
        FQE["<b>Federated Query Engine (FQE)</b>\nexecution against data sources · result  
assembly · lineage write"]
        DVL["<b>Data Visualization Language (DVL)</b>\nontology evaluation · deterministic  
chart contract selection"]
        NSA["<b>Narrative Synthesis Agent (NSA)</b>\npost-computation · anchored to result  
values · LLM call"]
        PAS["<b>Provenance Artifact Service (PAS)</b>\nassembles and seals Provenance  
Artifact from ALS records\nactive only for compliance-purpose queries"]
        LS["<b>Analytical Lineage Store (ALS)</b>\ncomputation provenance records\ntool call  
· SMR resolution · LQP · controls decision · execution record · narrative status"]
        Result["<b>MCP tool response</b>\ndisplay_spec + data + narrative + result_id\n+  
compliance block (if Provenance Artifact active)"]
    end

    subgraph Image["<b>Image/Chart Rendering</b> (Optional)"]
        vega2img["<b>vega2img</b> (optional)\nStandalone MCP render service · DVL → SVG /  
PNG\nRegistered directly with consumers – not part of Analytics Engine"]
    end

    subgraph dcs["Data Context Store (DCS)"]
        direction LR
        DCSMCP["<b>API/MCP Interface</b>\nMCP server runtime · tool/resource/prompt  
presentation · JWT validation"]
        SDR["<b>Semantic Data Repository (SDR)</b>\ndata models · object models · critical  
data elements\nquality rules · physical schemas · data lineage"]
        SMR["<b>Semantic Metrics Repository (SMR)</b>\nmetric definitions · dimensions ·  
hierarchies\naggregation rules · access policies · compliance metadata"]
        SMR --> SDR
        DCSMCP --> SDR & SMR
    end

```

```

subgraph entstore["Data Entitlements Store (DES) – External"]
    ENT[("<b>Data Entitlements Store (DES)</b>\nrole definitions · metric access sets ·  
dimension access sets\nrow scope templates · column masks\nmanaged at logical object and data  
element level")]
end

subgraph llmext["LLM Service (External)"]
    LLM[("<b>Language Model</b>\nIntent ranking · narrative synthesis\nCalled by IRA and  
NSA")]
end

subgraph backends["Data Sources"]
    SQL[("<b>SQL Warehouse</b>")]
    ODA[("<b>OpenData API</b>\nREST / OData")]
    GDA[("<b>Graph Data API</b>")]
end

ChatComp ~~~ CustomUI
CustomUI ~~~ Agents
SQL ~~~ ODA
ODA ~~~ GDA

Consumers -->|"JWT + structured MCP tool call"| MCP
Consumers -->|"render tool call (display_spec)"| Image
Consumers -->|"JWT + MCP tool call"| DCSMCP
MCP -->|"natural language query"| IRA
MCP -->|"structured tool call (bypass IRA) + JWT"| RAPL
IRA -->|"RAG retrieval"| SMR
IRA -->|"intent ranking"| LLM
IRA -->|"resolved request + JWT"| RAPL
RAPL -->|"role definition lookup"| ENT
RAPL -->|"entitlement projection (decisions + conditions)"| SVL
SVL -->|"Logical Query Plan (LQP)"| SCL
SCL -->|"controls decision record"| LS
SCL -->|"approved LQP"| PQP
PQP -->|"physical_mapping lookup"| SMR
PQP -->|"physical execution plan"| FQE
FQE --> backends
FQE -->|"execution record"| LS
FQE -->|"assembled result"| DVL
FQE -->|"assembled result"| NSA
NSA -->|"LLM call"| LLM
LS -->|"lineage records (compliance queries only)"| PAS
DVL -->|"DVL display spec"| Result
NSA -->|"governed narrative"| Result
PAS -->|"sealed compliance block"| Result

style analytics fill:#d9eaf7,stroke:#93c5fd

```

The Analytics Engine will be a single MCP server exposing three analytical tools (`run_analytics` , `list_operations` , `drilldown`) through a single MCP Capability Layer endpoint. It will contain exactly two bounded AI steps: the Intent Resolution Agent (IRA), which will identify the right governed operation from the user's natural language query, and the Narrative Synthesis Agent (NSA), which will summarise the computed result in plain text after execution. All stages between them — RAPL, SVL, SCL, PQP, and FQE — will be

entirely deterministic. The same resolved intent, access permissions, and data will always produce the same query plan, the same execution, and the same result.

For conversational consumers, the user's natural language query will be forwarded directly to the Analytics Engine. Intent resolution — selecting the right governed operation and binding parameters — will happen inside the engine's IRA. The Analytics Engine will return the display specification, structured data, and governed narrative; the AI Chat Platform will render the result. Structured API consumers (agents, custom UIs) may call `run_analytics` with an explicit `operation_id` and `params`, bypassing the IRA entirely.

The `vega2img` service will sit outside the Analytics Platform boundary as an optional, independently registered MCP render service. Consumers that cannot natively render the DVL display specification will register `vega2img` directly and call it as a separate tool invocation, passing the `display_spec` from the `run_analytics` response.

Component Summary

#	Component	Summary
1	AI Consumers	The AI Consumers layer is responsible for providing the external access points through which governed analytical requests reach the platform. It encompasses three consumer types: conversational AI platforms that mediate natural language queries, autonomous agents and scheduled pipelines that submit structured requests, and custom applications that call the platform via host-issued tokens. All three consumer types share a single governed entry point and a single controls pipeline; consumption mode affects only the caller's interaction pattern, not the trust model applied. For natural language queries, the consumer forwards the query and the caller's JWT to the Analytics Engine, which handles intent resolution internally. For structured requests, consumers supply an explicit operation identifier and parameters, bypassing intent resolution and routing directly to the entitlement layer. In all cases, the consumer receives a structured MCP tool response containing a display specification, result data, a governed narrative, and a lineage reference.
2	MCP Capability Layer (MCP)	The MCP Capability Layer (MCP) is responsible for providing the single governed entry point through which all AI consumers access the platform's analytical capabilities. It receives tool call requests over MCP Streamable HTTP transport, validates the inbound JWT, and routes the request to either the Intent Resolution Agent for natural language queries or directly to the Role-Aware Projection Layer for structured calls. Each exposed tool represents a bounded, named operation with a typed input schema and a governed execution path. There is no privileged or alternative execution path; all consumers receive the same controls-validated results regardless of how they access the platform. The MCP layer assembles and returns a structured tool response containing the DVL display specification, result data, governed narrative, lineage reference, and, where applicable, a sealed compliance block.

#	Component	Summary
3	Intent Resolution Agent (IRA)	The Intent Resolution Agent (IRA) is responsible for translating a natural language query into a structured, validated operation request. It receives the natural language query and the caller's JWT from the MCP Capability Layer and retrieves candidate operations from the SMR catalogue using embedding similarity search. A language model ranks the candidates, binds parameters to the leading operation, and derives a presentation preview indicating the anticipated chart type and axis structure. When the top candidate exceeds the confidence threshold, the resolved intent is forwarded directly to the Role-Aware Projection Layer. When intent is ambiguous, ranked candidate cards are returned to the consumer for selection or conversational refinement before execution proceeds. The IRA is the only AI step in the pre-computation pipeline and produces no output visible to the end user once intent is resolved.
4	Semantic Metrics Repository (SMR)	The Semantic Metrics Repository (SMR) is responsible for governing every analytical concept resolvable on the platform. It stores versioned, approved metadata definitions for metrics, dimensions, hierarchies, and analytical operations, and exposes them to the pipeline for resolution at query time. Every identifier in a request must be registered and approved in the SMR before it can be queried; the Semantic Validation Layer enforces this as an architectural constraint, not a configurable policy. Metric definitions declare formula, aggregation rules, data affinity, physical mappings, classification level, compliance relevance, and regulatory framework attributes. Operation and metric definitions are stored with embeddings to support the Intent Resolution Agent's retrieval. Authoring and approval flow through the Data Context Store's versioning and governance workflow, with Analytics Governance holding final approval authority over all definitions.
5	Role-Aware Projection Layer (RAPL)	The Role-Aware Projection Layer (RAPL) is responsible for computing the entitlement projection for every request before any query plan is compiled. It receives the resolved request and caller's JWT, retrieves role definitions from the Data Entitlements Store, and merges all active roles into a single entitlement profile. Against that profile it makes five categories of decision: data access clearance, metric access, dimension access, row scope, and result set column masking. Entitlement is conferred by role membership against governed logical concepts, not by database permissions, keeping policies stable as the underlying physical implementation changes. The completed entitlement projection, covering approved metrics and dimensions, resolved row scope conditions, and registered column masks, is passed to the Semantic Validation Layer for enforcement. Every entitlement decision is written to the Analytical Lineage Store as an audit record at query time.

#	Component	Summary
6	Semantic Validation Layer (SVL)	The Semantic Validation Layer (SVL) is responsible for validating the analytical request and compiling it into a platform-agnostic Logical Query Plan. It receives the entitlement projection from the Role-Aware Projection Layer together with the fully qualified analytical request and passes them through four sequential stages: well-formedness and SMR resolution, compliance signal evaluation, entitlement enforcement, and LQP generation. Every identifier is resolved against approved SMR definitions; any unregistered or unapproved identifier causes the pipeline to stop before a plan is compiled. Row scope filter nodes are injected and column masking directives are embedded as a top-level array in the plan, ensuring entitlement enforcement carries through to physical execution. The output is a Logical Query Plan expressed entirely in SMR-registered concepts, carrying no backend references, no SQL, and no physical schema identifiers. The SVL is entirely deterministic; no AI model runs inside it.
7	Semantic Controls Layer (SCL)	The Semantic Controls Layer (SCL) is responsible for applying five sequential checks to every query before releasing it to the Physical Query Planner. It receives the Logical Query Plan from the Semantic Validation Layer and evaluates it against the platform controls configuration. The five checks are: data scale (estimated scan volume against the configured row limit using SDR profiling statistics), complexity (LQP node count and join depth), classification gate (metric classification ceiling), compliance check (two-signal trigger for compliance-purpose queries), and concurrency (active query count against the platform limit). Every check must pass; there is no user, agent, or internal path that bypasses SCL evaluation. When all checks pass, the SCL assigns a timeout budget, writes a signed controls decision record to the Analytical Lineage Store, and releases the approved LQP to the Physical Query Planner.
8	Physical Query Planner (PQP)	The Physical Query Planner (PQP) is responsible for translating the controls-approved Logical Query Plan into backend-specific physical query fragments ready for execution. It receives the approved LQP from the Semantic Controls Layer and performs three operations in sequence: resolving the physical mapping for each metric node from the SMR definitions attached to the plan, grouping metric nodes by data affinity to produce one sub-plan per affinity group, and translating each sub-plan into the native query language of its target backend. Row scope filters, dimension filters, and column masking directives from the LQP are distributed to each sub-plan so that entitlement enforcement carries through to the physical layer. The output is a sub-plans array containing one entry per data affinity group, each carrying the backend identifier, dialect, column masks, and timeout budget. The PQP has no execution capability; it passes the sub-plans array to the Federated Query Engine.

#	Component	Summary
9	Federated Query Engine (FQE)	The Federated Query Engine (FQE) is responsible for executing physical sub-plans against registered backends and assembling the results. It is the only component in the platform with knowledge of execution backend connection details, including endpoints, credentials, and availability. It receives the sub-plans array from the Physical Query Planner, checks the result cache using the LQP signature, and on a cache miss routes each sub-plan to its registered backend by data affinity and capability. All sub-plans are executed concurrently and the FQE enforces the timeout budget assigned by the Semantic Controls Layer, handling partial results where one sub-plan times out while others complete. Sub-results from multiple backends are joined on shared dimensions and column masks are applied to the assembled result. The FQE writes a complete execution record to the Analytical Lineage Store and passes the assembled result to the Data Visualization Language and Narrative Synthesis Agent for presentation assembly.
10	Data Visualization Language (DVL)	The Data Visualization Language (DVL) is responsible for producing a deterministic display specification for every analytical result. It receives the assembled result from the Federated Query Engine and classifies it against a taxonomy of registered intent patterns. The ontology evaluator matches the result shape and intent classification against registered chart contracts in order of specificity, returning the highest-scoring match as the display specification. The AI model does not select chart types; the DVL makes the final binding decision, ensuring the same analytical pattern produces the same chart type across all users, sessions, and model versions. The TABLE_GOVERNED contract serves as an unconditional fallback, ensuring every query receives a valid display specification regardless of result shape. The intent pattern taxonomy and chart contract registry are initial sets, expected to be extended as the platform's visualisation vocabulary grows over time.
11	Narrative Synthesis Agent (NSA)	The Narrative Synthesis Agent (NSA) is responsible for producing a governed plain-language summary of each computed result. It runs in parallel with the Data Visualization Language after the Federated Query Engine has assembled the result, making a single tightly-scoped language model call anchored strictly to the computed values. Its prompt is constructed from the assembled result only: metric labels, row values, units, and dimension names; it is not given the user's original query or SMR governance context, preventing the narrative from interpreting or inferring beyond what was computed. Every numeric value in the narrative must be present in the result set, and a post-generation validation pass rejects the output if any value cannot be matched, with one regeneration attempt permitted. The NSA is optional and can be disabled via a platform configuration flag with no effect on computation, lineage, or display specification generation.

#	Component	Summary
12	Analytical Lineage Store (ALS)	The Analytical Lineage Store (ALS) is responsible for providing a complete, immutable record of how every analytical result was produced. It receives two writes per query: a controls decision record written by the Semantic Controls Layer before execution begins, and a full execution record written by the Federated Query Engine after execution completes. Each lineage record captures the original request, the SMR metric definition versions resolved, the entitlement projection in force, the controls decisions applied, the physical sub-plans executed, and the visualisation contract and narrative status. Records are written once and never mutated; corrections are made via new amendment documents that reference the original record. The store supports regulatory audit export via a filtered query API, with digitally signed export packages available for compliance review. Retention periods are configurable, with a platform default of seven years for query records.
13	Provenance Artifact Service (PAS)	The Provenance Artifact Service (PAS) is responsible for assembling and sealing a tamper-evident compliance record for queries that trigger the two-signal compliance classification. It is invoked in parallel with the Data Visualization Language and Narrative Synthesis Agent, but only when the Semantic Controls Layer determines that both the metric compliance flag and the SVL intent classification signal are active. It reads the controls decision record and execution record from the Analytical Lineage Store for the current query, assembles them into a Provenance Artifact document, and seals it by writing the artifact back to the ALS as an immutable sibling record. Export of the query result is blocked until sealing is confirmed, ensuring compliance-purpose results cannot leave the platform without an associated auditable record. The sealed compliance block is included in the MCP tool response and any party holding the platform's public key can independently verify the artifact has not been altered since sealing.

Request Flow

Diagram (mmdc unavailable)

```

sequenceDiagram
    autonumber
    participant C as AI Consumer
    participant MCP as API/MCP Interface
    participant IRA as Intent Resolution Agent
    participant LLM as Language Model
    participant RAPL as Role-Aware Projection Layer
    participant DES as Data Entitlements Store
    participant SVL as Semantic Validation Layer
    participant SMR as Semantic Metrics Repository
    participant SCL as Semantic Controls Layer
    participant ALS as Analytical Lineage Store
    participant PQP as Physical Query Planner
    participant FQE as Federated Query Engine
    participant BE as Data Sources
    participant DVL as Data Visualization Language
    participant NSA as Narrative Synthesis Agent
    participant PAS as Provenance Artifact Service
    participant vega2img as vega2img (optional)

    rect rgb(240, 255, 245)
        note over C,ALS: Single consumer request – full pipeline
        C->>MCP: run_analytics (NL query + JWT)
        MCP->>MCP: validate JWT signature · expiry · org claim

        MCP->>IRA: natural language query + JWT
        IRA->>SMR: vector similarity search (RAG)
        SMR-->>IRA: top-K candidate operations + metric definitions
        IRA->>LLM: candidate operations + user query (intent ranking prompt)
        note over IRA: Ranks candidates · binds params · scores confidence
        alt ambiguous intent
            IRA-->>MCP: confirmation card (requiresIntentConfirmation: true)
            MCP-->>C: confirmation card
            C->>MCP: confirmed: true + selected intent
            MCP->>IRA: confirmed intent
        end
    end

    IRA->>RAPL: resolved operation_id + params + JWT
    RAPL->>RAPL: validate JWT · extract role claims
    RAPL->>DES: retrieve role definitions
    DES-->>RAPL: metric access sets · dimension access sets · row scope templates ·

column masks
    note over RAPL: Merge role definitions · APPROVE/DENY per metric and dimension<br/>
    >Resolve row scope templates against JWT claims · register column masks
    RAPL->>SVL: entitlement projection (metric_access_set · dimension_access_set ·
    row_scope · column_masks)

    SVL->>SMR: resolve operation · metric IDs · dimension IDs
    SMR-->>SVL: definitions · aggregation rules · performance_impact_weight · compliance
metadata

    note over SVL: Stage 2 – Compliance signal evaluation<br/>Stage 3 – Enforce
    entitlement projection against request<br/>Stage 4 – LQP generation

    SVL->>SCL: Logical Query Plan (LQP)<br/>– no SQL · no backend refs · SMR concepts
    only

```

```

    note over SCL: data scale · complexity · classification · compliance ·
concurrency<br/>Blocks if any check fails

    SCL->>ALS: controls decision record (written before execution is invoked)
    SCL->>PQP: approved LQP

    note over PQP: Reads physical_mapping fields from metric nodes already in the LQP<br/>
>No SMR call required – all mappings carried in the plan

    note over PQP: Resolves physical_mapping per metric node<br/>Applies entitlement row
scope + column masks<br/>Compiles physical execution plan – no execution capability

    PQP->>FQE: physical execution plan

    FQE->>BE: query execution against data sources
    BE-->>FQE: raw result sets

    FQE->>ALS: execution record (data sources · latency · cache hit · column mask
applications)
    end

    rect rgb(255, 248, 240)
    note over FQE,NSA: Presentation assembly – parallel
    par
        FQE->>DVL: assembled result
        note over DVL: Ontology evaluation → deterministic chart contract selection<br/>
>AI does not select chart type
        DVL-->>MCP: DVL display specification
    and
        FQE->>NSA: assembled result
        note over NSA: Language model call<br/>Anchored strictly to result values ·
validation pass before inclusion
        NSA->>LLM: result summary prompt
        LLM-->>NSA: narrative text
        NSA-->>MCP: governed narrative (lead + detail + anchoredTo)
    and
        note over ALS,PAS: Only for compliance-purpose queries
        ALS->>PAS: lineage records (controls decision + execution record)
        note over PAS: Assembles Provenance Artifact from ALS records<br/>Seals artifact
– immutable from this point<br/>Blocks export until sealing confirmed
        PAS-->>MCP: sealed compliance block (regulatory_trace_id · triggered_frameworks ·
export_requires_lineage)
    end
    end

    MCP-->>C: display_spec + data + narrative + result_id + lineage_url<br/>+ compliance
block (if Provenance Artifact active)

    opt Consumer cannot natively render DVL specification
    C->>vega2img: render tool call (display_spec)
    vega2img-->>C: SVG / PNG
    end

```

The computation pipeline (RAPL → SVL → SCL → PQP → FQE) will be entirely deterministic and will contain no AI. The Analytical Lineage Store will receive two writes per query — a controls decision record before the

PQP is invoked and a full execution record after — ensuring the audit trail is complete regardless of whether execution succeeds.

Running example: This query is traced through every component below — each section's Example shows the same request at the next stage of the pipeline.

```
"Show me portfolio returns versus benchmark for my equity portfolios this quarter."
```

AI Consumers

The AI Consumers layer is responsible for providing the external access points through which governed analytical requests reach the platform. It encompasses three consumer types: conversational AI platforms that mediate natural language queries, autonomous agents and scheduled pipelines that submit structured requests, and custom applications that call the platform via host-issued tokens. All three consumer types share a single governed entry point and a single controls pipeline; consumption mode affects only the caller's interaction pattern, not the trust model applied. For natural language queries, the consumer forwards the query and the caller's JWT to the Analytics Engine, which handles intent resolution internally. For structured requests, consumers supply an explicit operation identifier and parameters, bypassing intent resolution and routing directly to the entitlement layer. In all cases, the consumer receives a structured MCP tool response containing a display specification, result data, a governed narrative, and a lineage reference.

Natural language path. When a user asks an analytical question, the consumer will forward the natural language query and the user's JWT to the Analytics Engine. The engine's IRA will handle operation selection, parameter binding, and if intent is ambiguous will return a confirmation card before proceeding to execution.

Structured path. Consumers that construct explicit `operation_id` + `params` payloads (agentic pipelines, custom analytics UIs, integration tests) will call `run_analytics` with structured arguments directly. The `list_operations` tool will return the entitled operation catalogue for consumers that build their own operation selection UI. Structured calls will bypass the IRA and route directly to the RAPL.

Response assembly. The conversation engine will render the DVL display specification inline. The `narrative` object in the response will contain the governed summary produced by the NSA. The `result_id` will be retained for any follow-up `drilldown` call or lineage inspection.

Example

The user's question will be relayed by the conversational AI directly to the Analytics Engine with the user's JWT. The consumer will not interpret, translate, or structure the query — it will forward it as-is.

```

POST /v1/mcp
Authorization: Bearer <host-issued-jwt>

{
  "jsonrpc": "2.0",
  "id": "req-8a3f2c",
  "method": "tools/call",
  "params": {
    "name": "run_analytics",
    "arguments": {
      "query": "Show me portfolio returns versus benchmark for my equity portfolios this quarter."
    }
  }
}

```

The Analytics Engine will process the request end-to-end and return a structured result. The conversation engine will render the grouped bar chart from the DVL display specification and surface the governed narrative as the assistant's reply.

MCP Capability Layer (MCP)

Governing principles: P2 — Controls before execution · P5 — Role-aware by default

The MCP Capability Layer (MCP) is responsible for providing the single governed entry point through which all AI consumers access the platform's analytical capabilities. It receives tool call requests over MCP Streamable HTTP transport, validates the inbound JWT, and routes the request to either the Intent Resolution Agent for natural language queries or directly to the Role-Aware Projection Layer for structured calls. Each exposed tool represents a bounded, named operation with a typed input schema and a governed execution path. There is no privileged or alternative execution path; all consumers receive the same controls-validated results regardless of how they access the platform. The MCP layer assembles and returns a structured tool response containing the DVL display specification, result data, governed narrative, lineage reference, and, where applicable, a sealed compliance block.

Tool Catalogue

The Analytics Engine will expose three tools. All analytical operations will be SMR-catalogue driven — the code will be the execution engine, not the operation registry. The SMR will own every operation definition: what parameters it needs, what metrics and dimensions it supports, and how deeply it runs through the pipeline via its `execution_profile`.

`run_analytics(operation_id: str, params: dict, jwt: str)` — Executes any SMR-registered operation. The operation's `execution_profile` in the SMR will determine which pipeline stages run.

`list_operations(domain: str | None, jwt: str)` — Returns the SMR operation catalogue with operation IDs, display names, required parameters, supported metrics/dimensions, and execution profiles. Only operations the authenticated user is entitled to execute will be returned.

`drilldown(result_id: str, hierarchy: str, selected_value: str | None, jwt: str)` — Navigates into a dimension hierarchy from a prior result. All filters, row scope conditions, and entitlement context from the original result will be preserved.

Execution Profiles

Each SMR operation will carry an `execution_profile` defined in its `analytical_operation` entry in the SMR catalogue. This will tell the pipeline executor which stages to invoke. No execution depth will be hardcoded in the MCP layer — it will always be determined by the SMR catalogue.

Profile	Pipeline stages
<code>data_retrieval</code>	Auth → IRA → RAPL → FQE → Lineage
<code>metric_query</code>	Auth → IRA → RAPL → SVL → SCL → PQP → FQE → Lineage
<code>full_analytical</code>	Auth → IRA → RAPL → SVL → SCL → PQP → FQE → DVL + NSA + PAS → Lineage

Intent Confirmation Cards

When intent is ambiguous, or when `requiresIntentConfirmation: true` is set on the operation, the IRA will return candidate cards before executing any query. The cards will be returned as the MCP response body in place of the analytical result. The consumer will render all candidates simultaneously; the user selects one, optionally refines it through the chat experience, then confirms to proceed.

Each card will include the resolved operation and parameters alongside a **presentation preview** — the anticipated chart type and axis structure — so the user can verify both what will be queried and how the result will be presented before execution commits.

The response payload will use a `candidates[]` array. A single-candidate response (governance override) will use the same structure with one entry:

```
{
  "intent_session_id": "ira-sess-20260605-wk4n",
  "candidates": [
    {
      "rank": 1,
      "confidence": 0.91,
      "operation_id": "compare_portfolios",
      "operation_label": "Compare Portfolios",
      "resolved_metrics": ["portfolio_return", "benchmark_return"],
      "resolved_dimensions": ["portfolio_id", "asset_class"],
      "time_period": "quarter_to_date",
      "filters": [{ "field": "asset_class", "operator": "eq", "value": "EQUITY" }],
      "preliminary_impact_estimate": 620,
      "classification": "INTERNAL",
      "presentation_hint": {
        "chart_type": "bar",
        "primary_dimension": "portfolio_id",
        "measures": ["portfolio_return", "benchmark_return"],
        "series_by": "metric"
      }
    },
    {
      "rank": 2,
      "confidence": 0.67,
      "operation_id": "portfolio_summary",
      "operation_label": "Portfolio Summary",
      "resolved_metrics": ["portfolio_return", "portfolio_nav"],
      "resolved_dimensions": ["portfolio_id"],
      "time_period": "quarter_to_date",
      "filters": [],
      "preliminary_impact_estimate": 280,
      "classification": "INTERNAL",
      "presentation_hint": {
        "chart_type": "table",
        "primary_dimension": "portfolio_id",
        "measures": ["portfolio_return", "portfolio_nav"],
        "series_by": null
      }
    }
  ],
  "action": "select or refine – re-submit with selected_candidate: <index> or a refinement message"
}
```

Selecting a candidate: re-submit `run_analytics` with `"selected_candidate": 0` (0-based index) to confirm the chosen interpretation and proceed to execution.

Refining through chat: send a natural language adjustment — the IRA will update the leading candidate's parameters and return a revised card set. Refinement turns will be bounded by `intentRefinementMaxTurns` (default 5) and recorded in the lineage record's `intent_session` field.

Capability Governance

Every capability invocation will pass through the full controls pipeline: input schema validation → capability availability check (feature flags and role entitlements) → Role-Aware Projection → Semantic Validation Layer → Semantic Controls Layer → Physical Query Planner → FQE → result assembly → lineage record write. A capability not enabled by a feature flag or accessible to the user's role will appear as `available: false` with a reason.

Example

A structured `run_analytics` tool call will arrive from the AI Chat Platform. The MCP Capability Layer will validate the JWT signature, confirm the token has not expired, and extract the claims. For a natural language query it will route to the Intent Resolution Agent; for a structured call it will route directly to the Role-Aware Projection Layer. The MCP Capability Layer will not interpret the parameters or make any analytical decisions; it will validate, route, and wait.

Intent Resolution Agent (IRA)

Governing principles: P2 — Controls before execution · P10 — Deterministic computation, not generation

The Intent Resolution Agent (IRA) is responsible for translating a natural language query into a structured, validated operation request. It receives the natural language query and the caller's JWT from the MCP Capability Layer and retrieves candidate operations from the SMR catalogue using embedding similarity search. A language model ranks the candidates, binds parameters to the leading operation, and derives a presentation preview indicating the anticipated chart type and axis structure. When the top candidate exceeds the confidence threshold, the resolved intent is forwarded directly to the Role-Aware Projection Layer. When intent is ambiguous, ranked candidate cards are returned to the consumer for selection or conversational refinement before execution proceeds. The IRA is the only AI step in the pre-computation pipeline and produces no output visible to the end user once intent is resolved.

Intent Resolution Pipeline

Diagram (mmdc unavailable)

```

flowchart LR
    NL["Natural language query\n(or refinement turn)"]
    RAG["RAG retrieval\nvector similarity search\nagainst SMR embeddings"]
    RANK["LLM intent ranking\ncandidate operations + query\n→ ranked candidates + bound  
params\n+ presentation preview"]
    CONF{"Ambiguous?"}
    CARDS["1-3 candidate cards\nreturned to consumer"]
    REFINE{"User tweaks\nor selects?"}
    OUT["Resolved intent\noperation_id + params + presentation_hint → RAPL"]

    NL --> RAG --> RANK --> CONF
    CONF -->|"yes"| CARDS
    CARDS --> REFINE
    REFINE -->|"selects a card"| OUT
    REFINE -->|"requests a change"| NL
    CONF -->|"no"| OUT

```

RAG Retrieval

At registration time, each `analytical_operation` and `analytical_metric` definition in the SMR will be embedded: the operation name, description, example phrasings, required parameters, and associated metric descriptions will be concatenated and encoded as a dense vector stored alongside the definition.

When a query arrives, the IRA will encode the natural language input and perform a vector similarity search against the SMR operation embeddings. The top-K candidate operations and their associated metric definitions will be retrieved. Only these candidates — not the full catalogue — will be injected into the LLM ranking prompt.

LLM Intent Ranking

The LLM will receive the top-K candidates and the user's query. It will rank candidates, bind parameters, and score confidence for each. It will also derive a `presentation_hint` for each candidate — the likely chart type and primary axes — based on the operation's result shape and the SMR operation definition. This preview will be included in every candidate card returned to the consumer.

If the top candidate's confidence score exceeds `intentConfidenceThreshold` (configurable, default 0.75) and leads the second candidate by more than `intentConfidenceBand` (configurable, default 0.1), the IRA will proceed directly to the RAPL with no card shown. Otherwise, up to three ranked candidate cards will be returned for the user to select or refine.

The LLM call will be constrained: the prompt will contain only the candidate operation definitions and the user's query. The LLM will have no access to result data, SMR governance metadata, or user entitlements — those will be enforced downstream by SVL and RAPL.

Multi-Candidate Selection

When intent is ambiguous, the IRA will return up to three ranked candidate cards in a single `candidates[]` array. The number of candidates returned will be determined by confidence clustering:

Situation	Cards shown
Top candidate above threshold, clear leader	0 — proceeds directly to RAPL
Top candidate below threshold, or top two within confidence band	2 candidates
Top three candidates within confidence band	3 candidates
<code>requiresIntentConfirmation: true</code> on the operation (governance override)	1 candidate — approval required regardless of confidence

The consumer will re-submit with `"selected_candidate": <index>` (0-based) to indicate which card the user chose. The IRA will then forward the selected candidate's resolved intent to the RAPL.

Conversational Refinement

After candidate cards are presented, the user may respond with a natural language adjustment rather than selecting a card. The IRA will treat the refinement as a constrained update: it will re-run the LLM with the selected or leading candidate as context and apply the requested changes to that candidate's parameters. An updated card set will be returned.

The loop will be bounded: a maximum number of refinement turns is configurable (`intentRefinementMaxTurns`, default 5). After the limit, the IRA will require the user to select from the current candidate set or start a new query. No data will be accessed and no query will be executed during the refinement loop — it will be entirely within the IRA's pre-execution scope. Each refinement turn will be recorded in the session context and included in the lineage record's `intent_session` field.

Presentation Preview

Each candidate card will include a `presentation_hint` block derived from the operation's result shape and the SMR operation definition.

Field	Description
<code>chart_type</code>	Predicted chart type — <code>bar</code> , <code>line</code> , <code>heatmap</code> , <code>scatter</code> , <code>table</code>
<code>primary_dimension</code>	The field that will appear on the X axis or as the primary grouping
<code>measures</code>	Metric fields that will appear as Y axis values or colour encoding
<code>series_by</code>	Dimension used to create series or colour bands (if applicable)

The `presentation_hint` will be a pre-execution estimate. The DVL will produce the authoritative display specification after execution, based on the actual result shape. The hint is informational only.

Structured API Path

Consumers that construct explicit `operation_id` + `params` (agentic pipelines, custom analytics UIs, integration tests) will call `run_analytics` with a structured payload directly. MCP will route these calls directly to the RAPL, bypassing the IRA. The `list_operations` tool will return the full entitled operation catalogue for consumers that build their own operation selection UI.

Example

Running example — portfolio manager asks:

"Show me portfolio returns versus benchmark for my equity portfolios this quarter."

The IRA will encode this query and retrieve the top-3 candidate operations from the SMR:

`compare_portfolios` (score 0.91), `portfolio_summary` (score 0.67), `benchmark_attribution` (score 0.61).

The top candidate exceeds the confidence threshold and leads by more than 0.1. The LLM will bind the resolved intent and derive a presentation preview:

```
{
  "operation_id": "compare_portfolios",
  "params": {
    "port
```

Confirm Analytical Query

INTERNAL

Compare Portfolios

Review the resolved intent before execution proceeds.

OPERATION	compare_portfolios
METRICS	portfolio_return · benchmark_return
DIMENSIONS	portfolio_id · asset_class
TIME PERIOD	Quarter to date
FILTER	asset_class = EQUITY
PERFORMANCE IMPACT	620 / 1,000 units
CLASSIFICATION	INTERNAL

Approving will execute the query. Rejecting returns you to the conversation.

Reject

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Confidence is 0.91 — above threshold, no candidate cards shown. Resolved intent will be forwarded to the RAPL.

Semantic Metrics Repository (SMR)

Governing principles: P1 — Semantic abstraction · P3 — Deterministic metric resolution · P9 — Administrator sovereignty

The Semantic Metrics Repository (SMR) is responsible for governing every analytical concept resolvable on the platform. It stores versioned, approved metadata definitions for metrics, dimensions, hierarchies, and analytical operations, and exposes them to the pipeline for resolution at query time. Every identifier in a request must be registered and approved in the SMR before it can be queried; the Semantic Validation Layer enforces this as an architectural constraint, not a configurable policy. Metric definitions declare formula, aggregation rules, data affinity, physical mappings, classification level, compliance relevance, and regulatory framework attributes. Operation and metric definitions are stored with embeddings to support the Intent Resolution Agent's retrieval. Authoring and approval flow through the Data Context Store's versioning and governance workflow, with Analytics Governance holding final approval authority over all definitions.

Concept Types

The SMR will hold at least the following three types of JSON metadata definition. The SMR is expected to be extensible in nature to accommodate other analytical concept types beyond those envisaged in this target state design:

Definition type	Description
<code>analytical_metric</code>	Governed metric definition — formula, aggregation, <code>data_affinity</code> , <code>physical_mapping</code> , <code>required_dimensions</code> , <code>performance_impact_weight</code> , <code>classification_level</code> , <code>compliance_relevant</code> , <code>regulatory_framework</code>
<code>analytical_dimension</code>	Governed dimension definition — <code>data_affinity</code> , <code>physical_mapping</code> , enumerated values or <code>hierarchical_flag</code> , <code>hierarchy_levels</code>
<code>analytical_operation</code>	Governed operation definition — <code>execution_profile</code> , <code>required_params</code> , <code>supported_metrics</code> , <code>supported_dimensions</code> , <code>default_visualization</code>

Metric Definition Schema

Every metric in the SMR will conform to the following schema. This is the authoritative reference for metric registration and validation:


```

{
  "id": "portfolio_return",
  "version": "2.1.0",
  "label": "Portfolio Return",
  "description": "Total return of a portfolio over the specified period, net of fees, expressed as
a percentage.",
  "formula": "(end_market_value - start_market_value + cash_flows) / start_market_value",
  "compliance_relevant": false,
  "regulatory_framework": [],
  "unit": "percentage",
  "aggregation": {
    "default": "value_weighted_average",
    "allowed": ["value_weighted_average", "equal_weighted_average", "sum"],
    "granularity": ["daily", "monthly", "quarterly", "annual", "since_inception"]
  },
  "dimensions": {
    "required": ["portfolio", "date"],
    "optional": ["asset_class", "currency", "benchmark"]
  },
  "data": {
    "domain": "portfolio",
    "sub_domain": "performance",
    "source_tables": ["fact_portfolio_daily", "dim_portfolio"],
    "refresh_cadence": "daily",
    "latency_sla": "T+1"
  },
  "governance": {
    "owner": "head_of_performance_analytics",
    "steward": "performance_analytics_team",
    "classification": "INTERNAL",
    "approved": true,
    "approved_by": "cdo_office",
    "approved_at": "2025-11-15T09:00:00Z",
    "effective_from": "2025-11-15",
    "deprecated": false
  },
  "lineage": {
    "upstream_metrics": [],
    "upstream_sources": ["positions_service", "pricing_service", "cash_flow_service"],
    "downstream_metrics": ["active_return", "information_ratio"]
  },
  "access": {
    "roles": ["portfolio_manager", "risk_officer", "analytics_governance"],
    "public": false
  },
  "display": {
    "format": "percentage",
    "decimals": 2,
    "sign_convention": "positive_is_gain",
    "benchmark_comparison": true
  }
}

```

All metric definitions will pass through a governance review and approval process before they are resolvable on the platform. A metric authored by a Metrics Modeller will not be queryable until it has been reviewed and approved by Analytics Governance.

Formula Language

The SMR formula language will express metric computation logic in terms of other registered metrics or canonical data source identifiers — never physical column names. This will decouple metric definitions from backend schema changes: renaming a physical table or column will require only a mapping update, not a metric definition change. The formula language will support arithmetic composition, conditional expressions, safe division with null protection, time-windowed aggregations, and filtered sub-aggregations.

SMR Authoring and Discovery

The SMR and SDR will be independent stores, both housed within the Data Context Store (DCS). The SMR will be a separate store for metric definitions; it will reference the SDR's data definitions but will not extend or depend on it structurally. The DCS will be the external container that holds both.

Metric authoring will use the DCS's native versioning and approval workflow. Metrics Modellers will author `analytical_metric`, `analytical_dimension`, and `analytical_operation` JSON metadata definitions through the DCS authoring interface; Analytics Governance will approve them before they become resolvable.

Discovery — AI models and agents will discover available operations by calling the `list_operations` MCP tool. `list_operations` will return operation IDs, display names, required parameters, supported metrics, supported dimensions, and execution profiles for all approved operations within the caller's entitlement scope. There will be no `smr://` MCP resource URIs and no separate `list_metrics`, `get_metric_definition`, `propose_metric`, or `approve_metric` MCP tools.

The Analytics Engine will query the SMR directly at request time — the SVL will resolve metric and operation definitions, and the FQE will resolve physical mappings. The DCS will also expose its own external API and MCP interface through which consumers and tooling can independently browse, inspect, and discover SMR metric definitions and SDR data definitions without going through the Analytics Engine.

Example

When the request reaches the SVL, the SMR will be the catalogue every identifier is resolved against. The SVL will ask the SMR to resolve the `compare_portfolios` operation, then resolve `portfolio_return` and `benchmark_return` as `analytical_metric` documents. The SMR will confirm both are approved for the `portfolio_manager` role and return their definitions, including `data_affinity ( portfolio )`, `required_dimensions ( portfolio_id , time_period )`, and `aggregation ( value_weighted_average )`. `asset_class` will resolve as an approved `analytical_dimension` document with an approved filter operator (`eq`). If either metric document were absent or not in `"status": "approved"` state, the SVL would return `METRIC_NOT_FOUND` and the pipeline would stop.

Role-Aware Projection Layer (RAPL)

Governing principles: P5 — Role-aware by default · P1 — Semantic abstraction

The Role-Aware Projection Layer (RAPL) is responsible for computing the entitlement projection for every request before any query plan is compiled. It receives the resolved request and caller's JWT, retrieves role definitions from the Data Entitlements Store, and merges all active roles into a single entitlement profile. Against that profile it makes five categories of decision: data access clearance, metric access, dimension access, row scope, and result set column masking. Entitlement is conferred by role membership against governed logical concepts, not by database permissions, keeping policies stable as the underlying physical implementation changes. The completed entitlement projection, covering approved metrics and dimensions, resolved row scope conditions, and registered column masks, is passed to the Semantic Validation Layer for enforcement. Every entitlement decision is written to the Analytical Lineage Store as an audit record at query time.

Entitlement policies will be managed in the **Data Entitlements Store (DES)** — an independent external component. Policies will be defined at the **logical object and data element level**: granting or restricting access to named metrics, dimensions, and data elements as governed concepts, never to physical tables, schemas, or column names. Projection will not be optional and will not be bypassable; every request will pass through RAPL, sitting between the IRA and the SVL.

Restriction Types

RAPL will make five categories of entitlement decision. Every decision will be made inside RAPL (Stage 5 of the projection lifecycle); the resulting conditions will be carried in the entitlement projection and enforced downstream — data and metric/dimension removals by the SVL during plan generation, row scope and column restrictions by the FQE at execution and result assembly.

Restriction type	Decided in RAPL	Enforced at
Data access	Stage 5 — data domain or classification ceiling not within the user's entitled scope is DENIED	SVL Stage 3 — request rejected before plan generation
Metrics access	Stage 5 — requested metric not in the entitled access set is DENIED	SVL Stage 3 — denied metric removed from plan; request rejected if a required metric is lost
Dimension access	Stage 5 — requested dimension not in the entitled access set is DENIED	SVL Stage 3 — denied dimension removed from plan
Row scope access	Stage 5–6 — population scope decided and resolved against JWT claims	PQP sub-plan generation — row scope filter injected into each sub-plan; enforced by FQE at execution
Result set column masking	Stage 5 — masked columns and masking mode registered	FQE result assembly — value replaced, redacted, or excluded

Projection Lifecycle

Diagram (mmdc unavailable)

```
flowchart LR
    START(["Fully qualified analytical\nrequest + JWT"])
    S1["**Stage 1**\nJWT Validation"]
    S2["**Stage 2**\nRole Claim\nExtraction"]
    S3["**Stage 3**\nDES Role Definition\nRetrieval"]
    S4["**Stage 4**\nMulti-Role\nMerge"]
    S5["**Stage 5**\nEntitlement\nDecisions"]
    S6["**Stage 6**\nRow Scope Template\nResolution"]
    S7["**Stage 7**\nEntitlement Projection\nOutput → SVL"]

    START --> S1 --> S2 --> S3 --> S4 --> S5 --> S6 --> S7
```

Stage 1 — JWT Validation. Validate the inbound JWT: signature, expiry, and org claim. **DENY** — request rejected immediately if any check fails. No further processing occurs on an invalid token.

Stage 2 — Role Claim Extraction. Extract the user's analytical role claims from the validated JWT using the configured `roleClaimField`. **DENY** — if no valid analytical role claims are present the request is rejected.

Stage 3 — DES Role Definition Retrieval. Look up the full role definition for each extracted role from the Data Entitlements Store (DES). Each definition declares the data access scope, metric access set, dimension access set, row scope templates, column masks, and classification ceiling for that role. **DENY** — if no valid role definitions are retrieved the request is rejected.

Stage 4 — Multi-Role Merge. Merge all retrieved role definitions into a single entitlement profile. Data, metric, and dimension access: union. Row scope: intersection (most restrictive wins). Column masks: union (masked by any role = masked for the user). No APPROVE/DENY decision is made here — this stage produces the profile against which all decisions are made in Stage 5.

Stage 5 — Entitlement Decisions. Five classes of decision will be made against the merged entitlement profile:

- **Data access** — the requested data domain and classification level will be checked against the user's entitled data access scope and classification ceiling. If either check fails the request will be **DENIED** (`DATA_NOT_ENTITLED`) before metric evaluation begins.
- **Metrics access** — each requested metric will be **APPROVED** or **DENIED** (`METRIC_NOT_ENTITLED`). Approval may be with or without dimensional constraints. If any required metric is denied the request will be rejected.
- **Dimension access** — each requested dimension will be **APPROVED** or **DENIED** (`DIMENSION_NOT_ENTITLED`). Entitlement will be evaluated per metric-dimension combination.
- **Row scope access** — each approved metric will carry an **APPROVED with population scope** decision: the row scope that defines which population of data the user is permitted to see.

- **Result set column masking** — each approved metric will carry an **APPROVED with column restrictions** decision where applicable.

No approved metric will reach the output without its full set of data access clearance, population scope, and column visibility conditions attached.

Stage 6 — Row Scope Resolution. Resolve the row scope templates from Stage 5 against the user's JWT claims. `{{user.claim_name}}` syntax will be expanded to concrete values at query time. **DENY** — if a required JWT claim for scope resolution is missing the request will be rejected.

Stage 7 — Entitlement Projection Output. Produce the completed entitlement projection: approved `data_access_scope`, approved `metric_access_set` (each with permitted aggregations and dimensional constraints), approved `dimension_access_set`, resolved `row_scope[]`, registered `column_masks[]`, and `classification_ceiling`. Write the full projection record to the ALS. Pass to SVL for Stage 3 enforcement.

Multi-Role Merge Strategy

Entitlement type	Merge strategy	Rationale
Data access	Union	Entitlement via any role is sufficient
Metrics access	Union	Entitlement via any role is sufficient
Dimension access	Union	Entitlement via any role is sufficient
Row scope access	Intersection (AND)	All scope conditions must be satisfied — most restrictive wins
Result set column masking	Union	A column masked by any role is masked for the user

Column Masking Modes

RAPL will support at least the following column masking modes. The platform is expected to be extensible in nature to facilitate other masking models beyond those envisaged in this target state design:

Mode	Column representation in result
<code>null_replacement</code>	Column value replaced with <code>null</code>
<code>redacted_label</code>	Column value replaced with <code>"[REDACTED]"</code>
<code>excluded</code>	Column omitted entirely from the result schema

Entitlement Audit

Every entitlement decision — data access clearance, approved and denied metrics, approved dimensions, row scope applied, and column masking registered — will be recorded in the Analytical Lineage Store (ALS) as part of the projection record. The record will capture the roles active at query time, each decision's APPROVE or DENY outcome and the basis for it, the resolved row scope for each approved metric, and the column restrictions in force.

Example

The RAPL will read the `portfolio_scope` claim from the JWT and resolve it against the `portfolio_manager` role's row scope template (`portfolio_id IN ({{user.portfolio_scope}})`):

```
{
  "data_access_scope":      "domain:portfolio · classification:INTERNAL",
  "row_scope": [
    { "dimension": "portfolio_id", "operator": "in",
      "values": ["GLOB_EQ_OPP", "UK_CORE_INC", "ASIA_PAC_GRW", "EUR_BAL_INC"] }
  ],
  "column_masks":          [],
  "classification_ceiling": "INTERNAL",
  "projection_basis":      "jwt_claim:portfolio_scope"
}
```

No column masks will apply — the `portfolio_manager` role has no masking rules for performance metrics. The row scope will be injected into the LQP as node `n3`. Any portfolio outside the four listed IDs will be excluded at the physical query level. The entitlement projection will be handed to the SVL.

Semantic Validation Layer (SVL)

Governing principles: P2 — Controls before execution · P10 — Deterministic computation, not generation

The Semantic Validation Layer (SVL) is responsible for validating the analytical request and compiling it into a platform-agnostic Logical Query Plan. It receives the entitlement projection from the Role-Aware Projection Layer together with the fully qualified analytical request and passes them through four sequential stages: well-formedness and SMR resolution, compliance signal evaluation, entitlement enforcement, and LQP generation. Every identifier is resolved against approved SMR definitions; any unregistered or unapproved identifier causes the pipeline to stop before a plan is compiled. Row scope filter nodes are injected and column masking directives are embedded as a top-level array in the plan, ensuring entitlement enforcement carries through to physical execution. The output is a Logical Query Plan expressed entirely in SMR-registered concepts, carrying no backend references, no SQL, and no physical schema identifiers. The SVL is entirely deterministic; no AI model runs inside it.

Four-Stage Validation Pipeline

Diagram (mmdc unavailable)

```

flowchart LR
    S1["**Stage 1**\nValid · Complete\nResolved · Compatible"]
    S2["**Stage 2**\nCompliance Signal\nEvaluation"]
    S3["**Stage 3**\nEntitlement\nEnforcement"]
    S4["**Stage 4**\nLQP Generation"]
    LQP(["Logical Query Plan (LQP)"])

    S1 --> S2 --> S3 --> S4 --> LQP

```

Stage 1 — Valid, Complete, Resolved & Compatible. The request must be well-formed, fully populated, and resolvable against the SMR before any further processing occurs. Required fields must be present and correctly typed. The `operation_id` must resolve to an approved `analytical_operation` metadata definition. Every metric ID must resolve to an approved `analytical_metric` metadata definition and every dimension ID to an approved `analytical_dimension` metadata definition. Params must conform to the operation's `required_params` schema. Cross-entity compatibility will be validated in the same pass. Anything unregistered, unapproved, missing, or incompatible will be rejected here — the pipeline will not proceed.

Stage 2 — Compliance Signal Evaluation. The SVL will combine two independent signals to determine the compliance disposition of the request. Signal 1 will be the IRA's compliance intent score — derived from the user's natural language query and forwarded as part of the resolved request. Signal 2 will be the `compliance_relevant` flag on each resolved `analytical_metric` metadata definition. If both signals are active the request will be escalated to the full compliance tier and the Provenance Artifact Service will be invoked. If the user's stated intent is compliance-driven but the requested metrics are not registered as compliance-relevant, the SVL will reject the request.

Stage 3 — Entitlement Enforcement. The SVL will apply the entitlement projection computed by the Role-Aware Projection Layer. Metrics and dimensions will be filtered to the caller's entitled scope. Row scope resolved by RAPL will be injected as scope filter nodes in the plan. Column masking directives from RAPL will be embedded in the LQP as a top-level `column_masks` array — carrying field name, masking mode, and the basis role for each masked column — so the Physical Query Planner can carry them through to the FQE for application to the assembled result. Any metric or dimension RAPL did not approve will be removed; if removal leaves the request without its required metrics the request will be rejected with an entitlement error rather than returning a partial result.

Stage 4 — LQP Generation. The validated, projected, and compliance-classified request will be compiled into a platform-agnostic Logical Query Plan — a directed acyclic graph of analytical operations expressed entirely in SMR-registered concepts. No SQL, no backend references, and no physical schema identifiers will appear in the LQP. Data affinity hints will be assigned per metric node to guide the Physical Query Planner's execution plan compilation. Column masking directives from Stage 3 will be included as a top-level `column_masks` array on the plan. A preliminary impact estimate (`preliminary_impact_estimate`) will be computed by summing the `performance_impact_weight` values of all resolved metric definitions and attached to the LQP. This is a Tier 1 coarse indicator of query weight available before row scope is fully known; the SCL will replace it with a precise scan-volume estimate at Check 1 time using SDR data profiling statistics.

Example

The SVL will receive the fully qualified analytical request together with RAPL's entitlement projection and pass them through all four stages. The output will be the Logical Query Plan passed to the Semantic Controls Layer:

```
{
  "lqp_id": "lqp-20260518-093243-r9xq",
  "intent_id": "int-20260518-093241-pk7m",
  "org_id": "acme-wealth",
  "nodes": [
    { "id": "n1", "type": "metric_scan",
      "metrics": ["portfolio_return", "benchmark_return"],
      "dimensions": ["portfolio_id"],
      "time_period": { "granularity": "quarterly", "anchor": "current" } },
    { "id": "n2", "type": "filter", "input": "n1",
      "predicate": { "field": "asset_class", "operator": "eq", "value": "EQUITY" } },
    { "id": "n3", "type": "row_scope", "input": "n2",
      "scope": { "field": "portfolio_id", "operator": "in",
        "values": ["GLOB_EQ_OPP", "UK_CORE_INC", "ASIA_PAC_GRW", "EUR_BAL_INC"] } },
    { "id": "n4", "type": "sort", "input": "n3",
      "field": "portfolio_return", "direction": "desc" }
  ],
  "output_node": "n4",
  "column_masks": [],
  "preliminary_impact_estimate": 620,
  "classification_required": "INTERNAL"
}
```

Node `n3` is the RAPL row scope — a first-class plan node, not a post-execution filter. `column_masks` is empty here because the `portfolio_manager` role carries no masking rules for performance metrics; for roles that do, the array would contain entries of the form `{ "field": "counterparty_id", "mode": "redacted_label", "basis_role": "risk_viewer" }`.

Semantic Controls Layer (SCL)

Governing principles: P2 — Controls before execution · P8 — Explainability at every layer · P9 — Administrator sovereignty

The Semantic Controls Layer (SCL) is responsible for applying five sequential checks to every query before releasing it to the Physical Query Planner. It receives the Logical Query Plan from the Semantic Validation Layer and evaluates it against the platform controls configuration. The five checks are: data scale (estimated scan volume against the configured row limit using SDR profiling statistics), complexity (LQP node count and join depth), classification gate (metric classification ceiling), compliance check (two-signal trigger for compliance-purpose queries), and concurrency (active query count against the platform limit). Every check must pass; there is no user, agent, or internal path that bypasses SCL evaluation. When all checks pass, the

SCL assigns a timeout budget, writes a signed controls decision record to the Analytical Lineage Store, and releases the approved LQP to the Physical Query Planner.

Controls Pipeline

Diagram (mmdc unavailable)

```
flowchart LR
    START(["Approved LQP"])
    S1["**1. Data Scale Check**"]
    S2["**2. Complexity Check**"]
    S3["**3. Classification Gate**"]
    S4["**4. Compliance Check**"]
    S5["**5. Concurrency Check**"]
    S6["**6. Timeout Budget**"]
    S7["**7. Controls Record & Release**\n→ PQP"]

    START --> S1 --> S2 --> S3 --> S4 --> S5 --> S6 --> S7
```

Check 1 — Data Scale Check. Using row counts, partition sizes, and time-series volume distributions held in the Semantic Data Repository, the SCL will compute a precise `estimated_scan_rows` value across all data sources in scope. This is the Tier 2 estimate — the final, authoritative scan-volume calculation — replacing the `preliminary_impact_estimate` carried in the LQP. It is the first point in the pipeline where full information is available: the LQP carries fully resolved row scope, and the SDR holds current data distribution statistics. The `estimated_scan_rows` value is compared against the `maxScanRows` limit in the controls configuration. A simple query spanning a large, unpartitioned fact table may exceed the limit and be blocked; a structurally complex query operating over a narrow time-window partition may pass comfortably. When blocked, the user will receive structured suggestions to narrow scope — reduce the time period, add a filter, or reduce the number of metrics.

Check 2 — Complexity Check. The SCL will evaluate the structural complexity of the LQP independently of data scale: total node count, join depth, and number of distinct data sources involved. Complexity limits protect the Physical Query Planner and the FQE from pathologically complex plans that could degrade execution performance regardless of underlying data volume. A query blocked by the complexity ceiling will receive an error identifying which dimension of complexity was exceeded.

Check 3 — Classification Gate. Every resolved metric carries a `data.classification` value set by the Metrics Modeller at registration. The SCL will identify the highest classification level across all metrics in the LQP and compare it against the permitted classification ceiling in the controls configuration. If any metric's classification exceeds the ceiling the query will be blocked. This check operates independently of entitlement enforcement — RAPL will have already confirmed the caller's access to each metric; the classification gate confirms the overall query's classification envelope is within the platform's operational boundary.

Check 4 — Compliance Check. A request will be classified as a compliance-type request when two independent signals are both active at evaluation time. When classified, the platform will invoke the

Provenance Artifact Service, apply framework-specific validation rules, and block result export until the artifact is sealed. Compliance relevance is declared on each metric at registration by the Metrics Modeller — the platform makes no compliance determination on a request without that declaration.

Provenance Artifact trigger — two signals, both required (AND logic)

Signal	Source	True when
Signal 1 — metric metadata	<code>compliance_relevant</code> field on <code>analytical_metric</code> SMR definition	At least one resolved metric has <code>compliance_relevant: true</code> . Set by the Metrics Modeller at registration.
Signal 2 — AI intent classification	SVL Stage 2 — Compliance Signal Evaluation	<code>compliance_purpose_score</code> $\geq$ <code>compliance_intent_threshold</code> (default 0.8, configurable). The SVL will classify the query's analytical intent — derived from the operation ID, resolved metric descriptions, and parameter values — and set <code>compliance_purpose: true</code> if the score meets the threshold.

When the Provenance Artifact is active, export of the result will be blocked until the artifact is confirmed written and sealed to the ALS. The `export_requires_lineage: true` flag in the response will signal this state to the consumer. The consumer will not present export affordances until the platform confirms sealing.

Framework-specific validation rules — additional parameter requirements, data constraints, lineage record types, and NSA output constraints — are declared exclusively on the metric definition in the SMR via the metric's `regulatory_framework` attribute, set by the Metrics Modeller at registration. The SCL will read and apply these rules directly from the resolved metric definitions at query time. No regulatory framework logic is hardcoded in the SCL.

Check 5 — Concurrency Check. The SCL will count the number of active queries currently in execution and compare it against the `maxConcurrentQueries` limit in the controls configuration. If the platform is at capacity, the query will be blocked with a `concurrency_limit_reached` response, and the caller will be advised to retry. This limit protects FQE coordination resources and ensures that no single burst of requests degrades response times across all active queries.

Step 6 — Timeout Budget Assignment. Once all five checks pass, the SCL will assign a `queryTimeoutSeconds` value to the query. This is not a blocking check — it is a budget allocation. The timeout is derived from the estimated scan volume and the configured per-engine timeout settings in the controls configuration. It will be attached to the controls output and forwarded with the LQP to the Physical Query Planner, which will pass it to the FQE for enforcement during execution.

Step 7 — Controls Record and Release. The SCL will write a signed controls decision record to the Analytical Lineage Store before forwarding the LQP to the Physical Query Planner. The record will capture the LQP identifier, the outcome of every check, the assigned timeout budget, and the compliance classification. Only after this record is confirmed written will the SCL release the query to the PQP. The lineage record is the authoritative confirmation that every control was applied.

Example

SCL will evaluate the LQP against the `acme-wealth` controls config:

Check	Value	Limit	Result
Data scale (estimated scan volume)	620M rows	1B rows	Pass
Complexity (node count)	4 nodes	50 nodes	Pass
Classification	INTERNAL	INTERNAL ceiling	Pass
Compliance	none triggered	—	Pass
Concurrency (active queries)	3	20	Pass

All checks will pass. SCL will write a controls decision record to the ALS before releasing to the PQP:

```
{
  "lqp_id": "lqp-20260518-093243-r9xq",
  "decision": "approved",
  "timestamp": "2026-05-18T09:32:44Z",
  "checks": ["data_scale_check", "complexity_check", "classification_gate", "compliance_check",
    "concurrency_check"],
  "result": "all_passed"
}
```

Physical Query Planner (PQP)

Governing principles: P1 — Semantic abstraction · P10 — Deterministic computation, not generation

The Physical Query Planner (PQP) will be the translation boundary between the logical and physical layers of the pipeline. It will receive the controls-approved Logical Query Plan from the SCL and produce a physical execution plan that the Federated Query Engine can execute against the data sources. Nothing above the PQP will have knowledge of physical schemas, table names, or backend-specific query representations. Nothing below it will operate on logical concepts. The same LQP will always produce the same physical execution plan: given identical metric definitions, filters, and time expressions, the PQP's output will be fully reproducible — a property required for lineage integrity.

Compilation Pipeline

Diagram (mmdc unavailable)

```

flowchart LR
    START(["Approved LQP"])
    S1["**1. physical_mapping Resolution**"]
    S2["**2. Entitlement Filter Application**"]
    S3["**3. Execution Plan Compilation**"]
    OUT(["Physical execution plan → FQE"])

    START --> S1 --> S2 --> S3 --> OUT

```

Step 1 — physical_mapping Resolution. For each `metric_scan` node in the LQP, the PQP will read the `physical_mapping` field from the resolved metric definition already attached to the node by the SVL. This field declares the registered data source identifier (`source`), the physical table or view (`table`), and the column or pre-computed measure (`measure`). No additional SMR call will be required — all physical mapping information will already be present in the LQP.

Step 2 — Entitlement Filter Application. Row scope filters and dimension filters from the LQP will be applied to each data source scope so that entitlement enforcement carries through to the physical layer. Column masking directives from the LQP's `column_masks` array will be carried forward in the execution plan for the FQE to apply to the assembled result.

Step 3 — Execution Plan Compilation. The PQP will compile the resolved metric nodes, filters, time range expansion, and sort/limit expressions into a physical execution plan ready for the FQE. The PQP will have no execution capability — it will not connect to data sources, manage timeouts, or assemble results. The FQE will own everything from that point forward.

Example

The PQP will receive the approved LQP for the portfolio manager query. Both `portfolio_return` and `benchmark_return` carry `data_affinity: "portfolio"` and `physical_mapping.source: "primary-warehouse"`. The PQP will read the physical table and measure references from the metric nodes, apply the RAPL row scope and asset class filter, expand `quarter_to_date` to the concrete date range `2026-04-01 → 2026-06-30`, and compile the physical execution plan for the FQE.

```

{
  "plan_id":          "plan-20260518-093244-m2kp",
  "lqp_id":           "lqp-20260518-093243-r9xq",
  "sources":          ["primary-warehouse"],
  "column_masks":     [],
  "query_timeout_seconds": 30
}

```

The physical execution plan will reference the resolved measure columns from `primary-warehouse`, the entitlement row scope filter, and the expanded date range. The FQE will execute this plan against the data source and return the assembled result. If the query also included a metric from a second data source, the PQP would include both sources in the execution plan and the FQE would handle the cross-source query.

Federated Query Engine (FQE)

Governing principles: P1 — Semantic abstraction · P4 — Complete analytical lineage · P10 — Deterministic computation, not generation

The Federated Query Engine (FQE) will be the only component in the platform with knowledge of execution backend connection details — endpoints, credentials, and availability. It will receive the physical execution plan from the Physical Query Planner, execute it against the data sources, assemble the results, and write a complete execution record to the lineage store. Execution plan compilation is the PQP's responsibility; the FQE will own everything from execution onward.

Execution Pipeline

Diagram (mmdc unavailable)

```
flowchart LR
    START(["Physical execution plan"])
    S1["**1. Plan Reception**"]
    S2["**2. Cache Check**"]
    CACHED(["Cached result returned"])
    S3["**3. Execution**"]
    S4["**4. Result Assembly**"]
    S5["**5. Result Caching**"]
    S6["**6. Lineage Record Writing**"]
    RESULT(["Assembled result + lineage record"])

    START --> S1 --> S2
    S2 -->|cache hit| CACHED
    S2 -->|cache miss| S3
    S3 --> S4 --> S5 --> S6 --> RESULT
```

Step 1 — Plan Reception. The FQE will receive the physical execution plan from the PQP and validate that every required data source is registered and available before proceeding.

Step 2 — Cache Check. The FQE will check the result cache using the LQP signature as the cache key. The key incorporates an entitlement hash derived from the caller's effective row scope and column masks, ensuring that two users with different entitlements will never be served each other's cached results. On a cache hit, the cached result is returned directly — steps 3–6 are skipped. Compliance-purpose queries bypass the cache and are always freshly executed.

Step 3 — Execution. The FQE will execute the physical execution plan against the registered data sources, enforcing the `queryTimeoutSeconds` budget assigned by the SCL. The FQE will support connectivity to data sources of at least the following types:

Data source type	Typical use
SQL warehouse or lakehouse	Primary performance, position, and risk data
Semantic layer	Pre-modelled governed metrics
REST / OpenData API	Reference data and third-party feeds
Graph data store	Relationship and counterparty data
OLAP engine	Pre-aggregated dimensional data

If a source times out while others complete, the FQE will assemble a partial result — representing missing metrics as null with a `timeout` provenance marker — and notify the user. If all sources time out, the query will fail and an execution record will be written with `timeout` status.

Step 4 — Result Assembly. Results from multiple data sources will be joined on shared dimensions. Column masks from the LQP's `column_masks` array will be applied to the assembled result before it is passed downstream.

Step 5 — Result Caching. The assembled result will be written to the result cache keyed by LQP signature. Results over 10 MB will bypass the cache and be streamed directly.

Step 6 — Lineage Record Writing. The FQE will write a complete execution record to the Analytical Lineage Store before returning the result — capturing data sources used, latency, cache status, any timeout events, and column mask applications.

Example

Both `portfolio_return` and `benchmark_return` have `data_affinity: "portfolio"`, so the FQE will execute the plan against the primary data source:

```
-- PQP physical execution plan received by FQE
SELECT
  p.portfolio_id,
  SUM(f.portfolio_return * f.market_value) / SUM(f.market_value) AS portfolio_return,
  SUM(f.benchmark_return * f.market_value) / SUM(f.market_value) AS benchmark_return
FROM fact_portfolio_daily f
JOIN dim_portfolio p ON f.portfolio_id = p.portfolio_id
WHERE p.asset_class = 'EQUITY'
  AND f.portfolio_id IN ('GLOB_EQ_OPP', 'UK_CORE_INC', 'ASIA_PAC_GRW', 'EUR_BAL_INC')
  AND f.date BETWEEN '2026-04-01' AND '2026-06-30'
GROUP BY p.portfolio_id
ORDER BY portfolio_return DESC
```

Assembled result:

```
{
  "result_id": "res-20260518-093247-wk4n",
  "latency_ms": 1187,
  "backends_used": ["primary-warehouse"],
  "rows": [
    { "portfolio_id": "GLOB_EQ_OPP", "portfolio_return": 4.21, "benchmark_return": 3.85 },
    { "portfolio_id": "ASIA_PAC_GRW", "portfolio_return": 3.67, "benchmark_return": 3.90 },
    { "portfolio_id": "UK_CORE_INC", "portfolio_return": 2.87, "benchmark_return": 2.54 },
    { "portfolio_id": "EUR_BAL_INC", "portfolio_return": 1.93, "benchmark_return": 2.31 }
  ]
}
```

The FQE will write an execution record to the Analytical Lineage Store (ALS) and pass the assembled result in parallel to the DVL and NSA.

Data Visualization Language (DVL)

Governing principles: P7 — Deterministic visualisation

The Data Visualization Language (DVL) is responsible for producing a deterministic display specification for every analytical result. It receives the assembled result from the Federated Query Engine and classifies it against a taxonomy of registered intent patterns. The ontology evaluator matches the result shape and intent classification against registered chart contracts in order of specificity, returning the highest-scoring match as the display specification. The AI model does not select chart types; the DVL makes the final binding decision, ensuring the same analytical pattern produces the same chart type across all users, sessions, and model versions. The TABLE_GOVERNED contract serves as an unconditional fallback, ensuring every query receives a valid display specification regardless of result shape. The intent pattern taxonomy and chart contract registry are initial sets, expected to be extended as the platform's visualisation vocabulary grows over time.

Intent Pattern Taxonomy

Intent patterns are registered classifications in the DVL. The initial set is defined below; new patterns will be added as the platform's analytical vocabulary grows.

Intent pattern	Description	Typical trigger phrases
COMPARISON	Comparing a metric across discrete categories	"compare", "by", "across", "ranked by", "top N"
TREND	Showing how a metric changes over time	"over time", "trend", "history", "30-day", "since"
DISTRIBUTION	Showing the spread or concentration of a metric	"distribution", "breakdown", "composition", "concentration"

Intent pattern	Description	Typical trigger phrases
THRESHOLD	Comparing a metric against a limit or benchmark	"exceeding", "within limit", "versus benchmark", "breach"
ATTRIBUTION	Decomposing a metric into contributing factors	"attribution", "contribution", "drivers", "breakdown by"
RELATIONSHIP	Showing correlation or dependency between metrics	"versus", "correlation", "scatter", "risk-return"
COMPOSITION	Showing part-to-whole relationships	"proportion", "weight", "allocation", "share"

Chart Contract Table

Chart contracts are registered specifications in the DVL, each binding an intent pattern to a chart type, axis assignments, and interaction semantics. The initial set is defined below; new contracts will be registered as the platform's visualisation vocabulary is extended.

Contract name	Intent patterns matched	Chart type	Key axis assignments	Interaction semantics
BAR_MULTI_SERIES_COMPARISON	COMPARISON	Bar	X: primary categorical dimension (sorted by primary metric DESC); Y: metric value; Colour: metric series or secondary dimension	Click: drilldown; Hover: tooltip; Selection: multi-point
LINE_TIME_SERIES_TREND	TREND	Line	X: temporal dimension; Y: metric value; Colour: metric series; Reference line: injected if <code>compare_to</code> present	Hover: crosshair tooltip; Click data point: surface lineage; Brush: zoom X-axis
HEATMAP_THRESHOLD_MATRIX	THRESHOLD	Heatmap	X: first categorical dimension; Y: second categorical dimension; Colour: metric as % of threshold (diverging scale, midpoint at 100% of limit)	Click cell: drilldown into dimension intersection

Contract name	Intent patterns matched	Chart type	Key axis assignments	Interaction semantics
TREEMAP_COMPOSITION	COMPOSITION , DISTRIBUTION	Treemap	Area: proportional to metric value; Colour: secondary metric (diverging scale); Label: dimension value + metric value	Click tile: drilldown to next hierarchy level; Hover: tooltip with all metrics
WATERFALL_ATTRIBUTION	ATTRIBUTION	Waterfall	X: contribution dimension; Y: contribution value (positive/negative); Colour: positive (green), negative (red), total (grey)	Hover: contribution value and percentage
SCATTER_RISK_RETURN	RELATIONSHIP	Scatter	X: first metric (conventionally risk); Y: second metric (conventionally return); Colour: categorical dimension; Size: optional third metric	Hover: all metric values; Reference lines: quadrant boundaries from benchmark values if present
TABLE_GOVERNED	Fallback (any)	Table	All result columns; column labels from SMR; inline sparklines for temporal metrics	Column sorting; column filtering; export to CSV and JSON

Override Mechanism

Power Analysts will be able to override the ontology's chart selection for a single result by expressing an explicit chart type preference in their query. Overrides will be subject to the requested chart type being in the platform's configured `allowedChartTypes` list and the result schema being compatible with the requested chart type. Incompatible overrides will be rejected with an explanation. All overrides will be logged in the lineage record as analyst-requested deviations from the governing ontology.

Example

The ontology evaluator will classify the result: two metrics across four named entities, with a natural comparison relationship between return and benchmark. This will match the `COMPARISON` pattern. The highest-scoring contract will be `BAR_MULTI_SERIES_COMPARISON` :

```
{
  "display_spec_id": "dsp-20260518-093248-f3xp",
  "contract": "BAR_MULTI_SERIES_COMPARISON",
  "intent_pattern": "COMPARISON",
  "chart_type": "bar",
  "encoding": {
    "x": { "field": "portfolio_id", "type": "nominal" },
    "y": { "field": "value", "type": "quantitative" },
    "color": { "field": "metric", "type": "nominal" },
    "xOffset": { "field": "metric", "type": "nominal" }
  },
  "title": "Portfolio Return vs Benchmark — Q2 2026"
}
```

Narrative Synthesis Agent (NSA)

Governing principles: P6 — Governed narrative · P10 — Deterministic computation, not generation

The Narrative Synthesis Agent (NSA) is responsible for producing a governed plain-language summary of each computed result. It runs in parallel with the Data Visualization Language after the Federated Query Engine has assembled the result, making a single tightly-scoped language model call anchored strictly to the computed values. Its prompt is constructed from the assembled result only: metric labels, row values, units, and dimension names; it is not given the user's original query or SMR governance context, preventing the narrative from interpreting or inferring beyond what was computed. Every numeric value in the narrative must be present in the result set, and a post-generation validation pass rejects the output if any value cannot be matched, with one regeneration attempt permitted. The NSA is optional and can be disabled via a platform configuration flag with no effect on computation, lineage, or display specification generation.

Anchoring and Validation

The NSA will produce three output fields:

Field	Description
<code>narrative.lead</code>	A single sentence summarising the top-level finding
<code>narrative.detail</code>	Two to four sentences covering the most significant data points, one per dimension value or notable comparison
<code>narrative.anchoredTo</code>	An array of dimension value identifiers from the result that the narrative references — used for post-generation validation

After generation, the NSA will run a validation pass: every numeric value in the narrative must be present in the result set. If any value cannot be matched to a result row, the narrative will be rejected and a single regeneration will be attempted. If the second attempt also fails validation, the `narrative` field will be omitted from the response and the failure will be recorded in the lineage record.

Model Selection

Query type	Model	Rationale
Standard queries (≤ 5 metrics, ≤ 3 dimensions)	Standard language model	Low latency; sufficient for concise, anchored summaries
Complex queries (attribution, multi-portfolio, regulatory)	Extended language model	Richer result structure requires more precise prose

The model used will be recorded in `narrative_status` in the lineage record.

Feature Flag

Narrative synthesis will be controlled by the `features.narrativeSynthesis` platform configuration flag. When disabled, the NSA will not be invoked and no `narrative` field will be included in the response. The default is enabled. Disabling narrative synthesis will have no effect on computation, lineage, or display spec generation.

Example

The NSA will receive the assembled result and produce:

```
{
  "narrative": {
    "lead": "2 of your 4 equity portfolios outperformed their benchmark this quarter.",
    "detail": "Global Equity Opportunities returned 4.21% against a benchmark of 3.85%. UK Core Income returned 2.87% against its benchmark of 2.54%. Asia Pacific Growth and EUR Balanced Income underperformed, returning 3.67% and 1.93% respectively against benchmarks of 3.90% and 2.31%.",
    "anchoredTo": ["GLOB_EQ_OPP", "UK_CORE_INC", "ASIA_PAC_GRW", "EUR_BAL_INC"]
  }
}
```

Post-generation validation will confirm every verbatim numeric value cited in the narrative is present in the assembled result rows. Residual hallucination risk applies to non-literal claims such as proportional expressions.

Analytical Lineage Store (ALS)

Governing principles: P4 — Complete analytical lineage · P8 — Explainability at every layer

The Analytical Lineage Store (ALS) is responsible for providing a complete, immutable record of how every analytical result was produced. It receives two writes per query: a controls decision record written by the Semantic Controls Layer before execution begins, and a full execution record written by the Federated Query Engine after execution completes. Each lineage record captures the original request, the SMR metric definition versions resolved, the entitlement projection in force, the controls decisions applied, the physical

sub-plans executed, and the visualisation contract and narrative status. Records are written once and never mutated; corrections are made via new amendment documents that reference the original record. The store supports regulatory audit export via a filtered query API, with digitally signed export packages available for compliance review. Retention periods are configurable, with a platform default of seven years for query records.

Storage Design

Lineage records will be stored in an object store — one JSON document per query at key `lineage/{org_id}/{yyyy}/{mm}/{dd}/{result_id}.json`. Records will be write-once and never mutated. Post-hoc compliance annotations will be written as sibling documents (`{result_id}_amendment_{n}.json`) referencing the original `result_id`.

A thin relational database search index will hold only scalar fields required for filtered search queries. The full record will always be fetched from the object store; the index will never be the source of truth for record content.

Per-Query Stored Elements

Element	Storage	Content
Lineage record	Object store — <code>lineage/{org_id}/{yyyy}/{mm}/{dd}/{result_id}.json</code>	Complete chain: tool call parameters → SMR resolution → projection record → LQP → controls decision → FQE execution record → result schema → visualisation contract → narrative synthesis status
SMR snapshot	Embedded in lineage record (<code>resolved_metrics</code>)	For each metric in the query: metric ID, SMR definition version at query time
Projection record	Embedded in lineage record	Roles, requested metrics, projected metrics, blocked metrics, row scope, column masks
FQE execution record	Embedded in lineage record (<code>execution</code>)	Data sources used, queries executed, latencies, scan volume, cache hit status
Controls decision	Embedded in lineage record (<code>controls_decision</code>)	Threshold decisions — data scale, complexity, classification, compliance, concurrency — including blocked queries
Search index row	Relational database <code>analytics.lineage_index</code>	Scalar fields for filtered search — <code>result_id</code> , <code>org_id</code> , <code>user_sub</code> , <code>regulatory_frameworks</code> , <code>error_code</code> , <code>cache_hit</code> , <code>created_at</code> , <code>expires_at</code>
Result artefact	Object storage	CSV result set, chart SVG, narrative text — stored per query

Search Index DDL: `analytics.lineage_index`

```
CREATE TABLE analytics.lineage_index (  
  result_id      TEXT      PRIMARY KEY,  
  org_id         TEXT      NOT NULL,  
  user_sub       TEXT      NOT NULL,  
  regulatory_frameworks TEXT,  
  error_code     TEXT,  
  cache_hit      BOOLEAN   NOT NULL,  
  created_at     TIMESTAMPTZ NOT NULL,  
  expires_at     TIMESTAMPTZ NOT NULL  
);
```

Data Isolation

Every lineage document will be stored under an `org_id`-prefixed key in the object store, and every row in the `analytics.lineage_index` table will carry an `org_id` column. Row-Level Security (RLS) in the relational database will enforce access isolation on the index table. Object store access will be gated by the platform's API, which will validate the JWT `org_id` claim before resolving any object key.

Retention

Rule	Specification
Query records	Platform default: 2,555 days (7 years) . Configurable.
Lineage records	Retained at least as long as the corresponding query record. Cannot be deleted independently.
SMR metric versions	Retained indefinitely — metric version history must be preserved for lineage reconstruction.
Controls events	Retained at least as long as query records.
Result artefacts (object storage)	Default: 365 days. Configurable. Lineage record references are preserved even after object storage expiry.
Blocked queries	Retained in full — queries that fail governance checks are as important to retain as successful ones.

Immutability

Lineage records will never be modified after writing. There will be no update or delete path available to any user — including Platform Admin. Corrections to erroneous records will produce new records that reference the original via a `supersedes` relationship. This constraint will be enforced at the database layer, not only by application logic.

Regulatory Audit Export

```
GET /v1/audit/queries
  ?user_id={user_id}
  &from={ISO8601}
  &to={ISO8601}
  &metric_ids[]={metric_id}
  &include_lineage=true
  → Returns all queries matching the filter with full lineage records
```

Audit export packages will include query records with timestamps and user identifiers, lineage records with metric definition versions, controls decisions, role-aware projection records showing entitlements in force at query time, and result artefacts within the retention period. All export packages will be digitally signed by the platform using the platform signing key registered at deployment.

Example

Two lineage writes will occur for each query — one before execution (the controls decision record) and one after (the full execution record):

```
{
  "result_id":      "res-20260518-093247-wk4n",
  "org_id":         "acme-wealth",
  "user_sub":       "idp|user_xyz",
  "lqp_id":         "lqp-20260518-093243-r9xq",
  "cache_hit":      false,
  "controls_decision": {
    "approved": true,
    "checks_passed": ["data_scale_check", "complexity_check", "classification_gate",
"compliance_check", "concurrency_check"]
  },
  "sub_plans": [
    {
      "backend": "primary-warehouse",
      "dialect": "sql",
      "latency_ms": 1187,
      "row_count": 4,
      "cache_hit": false
    }
  ],
  "resolved_metrics": [
    { "metric_id": "portfolio_return", "version": "2.1.0" },
    { "metric_id": "benchmark_return", "version": "1.4.2" }
  ],
  "projection_record": {
    "roles":      ["portfolio_manager"],
    "row_scope":  [{ "field": "portfolio_id", "operator": "in",
"values": ["GLOB_EQ_OPP", "UK_CORE_INC", "ASIA_PAC_GRW", "EUR_BAL_INC" ]}],
    "column_masks": []
  },
  "visualisation": { "contract": "BAR_MULTI_SERIES_COMPARISON" },
  "narrative_status": { "generated": true, "validation": "passed" },
  "created_at":      "2026-05-18T09:32:47Z",
  "expires_at":      "2033-05-18T09:32:47Z"
}
```

The document will be immutable from the moment of writing. The full chain — original query, controls decision, metric definition versions, entitlements, physical backend call, and chart contract — will be retrievable from the object store under `result_id`.

Provenance Artifact Service (PAS)

Governing principles: P4 — Complete analytical lineage · P2 — Controls before execution · P9 — Administrator sovereignty

The Provenance Artifact Service (PAS) is responsible for assembling and sealing a tamper-evident compliance record for queries that trigger the two-signal compliance classification. It is invoked in parallel with the Data Visualization Language and Narrative Synthesis Agent, but only when the Semantic Controls Layer determines that both the metric compliance flag and the SVL intent classification signal are active. It reads the controls decision record and execution record from the Analytical Lineage Store for the current query,

assembles them into a Provenance Artifact document, and seals it by writing the artifact back to the ALS as an immutable sibling record. Export of the query result is blocked until sealing is confirmed, ensuring compliance-purpose results cannot leave the platform without an associated auditable record. The sealed compliance block is included in the MCP tool response and any party holding the platform's public key can independently verify the artifact has not been altered since sealing.

Purpose

The Provenance Artifact will be a sealed, tamper-evident record that satisfies regulatory requirements for demonstrable audit trail on compliance-purpose queries. It will be distinct from the standard lineage record that every query produces. The lineage record will be the full computation trace. The Provenance Artifact will be the governed, signed output of that trace — structured for regulatory consumption, explicitly linked to the frameworks that triggered it, and sealed before the result is returned to the consumer.

Assembly and Sealing

The PAS will read the controls decision record and execution record that the ALS has written for the current query. It will assemble them into a Provenance Artifact document — adding regulatory trace identifiers, framework tags, and the compliance intent score — and seal it by writing the sealed artifact back to the ALS as a sibling document (`{result_id}_provenance.json`). The artifact will be immutable from the moment of sealing. No further write or amendment will be permitted without creating a new amendment document referencing the original.

Export Gate

Until sealing is confirmed, export of the query result will be blocked. The PAS will set `export_requires_lineage: true` in the compliance block it returns to the MCP layer. The consumer will not present export affordances until the platform confirms sealing is complete.

Compliance Block Structure

Field	Description
<code>compliance_purpose</code>	<code>true</code> — confirms the two-signal trigger was active for this query
<code>intent_score</code>	The <code>compliance_purpose_score</code> from SVL
<code>triggered_by_metrics</code>	Metric IDs whose <code>compliance_relevant: true</code> flag contributed Signal 1
<code>triggered_by_frameworks</code>	Regulatory framework identifiers from the triggered metrics' SMR definitions
<code>regulatory_trace_id</code>	The trace record identifier written to the ALS regulatory partition for the triggered framework(s)

Field	Description
<code>artifact_set_version</code>	Artifact schema version — used by regulatory consumers to validate the structure
<code>export_requires_lineage</code>	<code>true</code> — consumer must not present export until sealing is confirmed
<code>classification_ceiling_applied</code>	<code>true</code> if a resolved metric triggered the RESTRICTED classification ceiling

Example

For a regulatory query where the resolved metrics carry `compliance_relevant: true` and regulatory framework attributes in their SMR definitions, and the SVL's compliance intent score is `0.94`, both signals will be active. The PAS will assemble the Provenance Artifact, seal it in the ALS, and return:

```
{
  "compliance_purpose":      true,
  "intent_score":           0.94,
  "triggered_by_metrics":   ["<metric_id_a>", "<metric_id_b>"],
  "triggered_by_frameworks": ["<framework_id>"],
  "regulatory_trace_id":    "trace-20260518-093247-<framework_id>",
  "artifact_set_version":   "1.0",
  "export_requires_lineage": true,
  "classification_ceiling_applied": true
}
```

This block will be included in the MCP tool response under the `compliance` key. The consumer will render an appropriate disclosure to the user and withhold export affordances until the platform signals sealing is complete.

MCP Response Format

The Analytics Engine will be headless: it will produce no rendered output. Every successful analytical request will return a structured MCP tool response. The response will contain a DVL display specification, structured result data, an optional governed narrative, a lineage reference, and — when the compliance trigger is active — a sealed compliance block.

DVL Specification Format

DVL will use a discriminated JSON envelope with two types:

Chart specification (`type: "chart"`) — produced when a chart contract matches:

```
{
  "type": "chart",
  "mark": "bar",
  "data": { "values": [ { "portfolio": "Global Equity", "tracking_error": 0.042 }, ... ] },
  "encoding": {
    "x": { "field": "portfolio", "type": "nominal", "title": "Portfolio" },
    "y": { "field": "tracking_error", "type": "quantitative", "title": "Tracking Error" }
  },
  "colorScheme": ["#003f5c", "#58508d", "#bc5090"],
  "formatHints": {
    "tracking_error": { "format": ".2%", "unit": "%" }
  }
}
```

Table specification (`type: "table"`) — produced when no chart contract matches:

```
{
  "type": "table",
  "columns": [
    { "field": "portfolio", "label": "Portfolio", "type": "string" },
    { "field": "tracking_error", "label": "Tracking Error", "type": "number", "format": ".2%" },
    { "field": "limit", "label": "Limit", "type": "number", "format": ".2%" },
    { "field": "breached", "label": "Breached", "type": "boolean" }
  ],
  "data": [ ... ],
  "thresholds": [
    { "field": "tracking_error", "operator": "gt", "reference": "limit", "severity": "warning" }
  ]
}
```

Column labels in table specifications will come from SMR metric and dimension `display.label` values — never from physical field names.

External Components

The following components will appear in the architecture diagram and interact with the Analytics Engine but will sit outside its boundary. They will not be built or owned by the Analytics Engine; they will be pre-existing or independently deployed services that the platform integrates with.

Conversational AI — Chat Front End

The AI Chat Platform will be the conversational consumer of the Analytics Engine. It will relay natural language questions from users to the Analytics Engine and render the structured results it receives. Intent resolution — identifying which governed operation matches the user's question and binding its parameters — will be performed inside the Analytics Engine by the IRA. The AI Chat Platform will perform no NL translation and will have no dependency on the SMR operation catalogue.

The AI Chat Platform will forward the user's natural language query and JWT to the Analytics Engine via `run_analytics`. If the Analytics Engine returns a confirmation card, the AI Chat Platform will render it to the

user and re-submit with `confirmed: true` when the user approves. It will render the DVL `display_spec` inline, surface the governed `narrative` as the assistant's reply, and retain the `result_id` for follow-up `drilldown` calls.

The AI Chat Platform will have no access to physical schemas, execution backends, or metric definitions. Entitlement enforcement, intent resolution, query planning, and execution will be entirely the Analytics Engine's responsibility.

Semantic Data Repository (SDR)

The Semantic Data Repository will be a pre-existing organisational component — the governed store of JSON-based data metadata definitions that describe the organisation's information assets. It will exist independently of the Analytics Platform and will not be built or owned by it. For most organisations it will already exist before the Analytics Platform is deployed.

The SDR will contain the organisation's foundational data context: data models, object models, critical data elements, quality rules, physical schemas, and data lineage records — *what data exists and how it is structured*. The SMR will be a separate store for metric metadata definitions — *what the data means analytically* and how it should be calculated, aggregated, and governed. Both will be independent stores housed within the Data Context Store (DCS).

The `physical_mapping` fields in SMR metric definitions will resolve against SDR schema metadata to identify the physical tables and columns that back each metric. The DCS search index (spanning both SMR and SDR) will support the `list_operations` tool and the IRA's vector similarity search over SMR operation and metric embeddings.

Data Entitlements Store (DES)

The Data Entitlements Store (DES) will be an independent external component that holds the organisation's data and analytics entitlement policies. It will be managed separately from the Analytics Engine and from the data platform — a dedicated governance control point.

Entitlement policies will be declared at the **logical object and data element level**. A policy will grant or restrict access to a named metric, a named dimension, or a named data element as governed concepts. Policies will not reference physical tables, schemas, column names, or connection strings. This separation will ensure entitlements remain stable as the underlying physical implementation evolves, and remain comprehensible to business data owners, compliance teams, and governance teams who have no visibility into the data platform's internal structure.

RAPL will read role definitions from the DES at query time, keyed on the role claims extracted from the caller's JWT. Changes to entitlement policies will not require changes to metric definitions, platform configuration, or backend schemas.

vega2img

`vega2img` will be an optional, independently deployed MCP render service. It will convert the Analytics Engine's DVL display specification into a static image — SVG or PNG — for consumers that cannot natively render a DVL specification inline.

`vega2img` will be deployed and registered independently of the Analytics Engine, registered directly with the AI consumer as a peer MCP server. The consumer will call it as a separate tool invocation, passing the `display_spec` returned by the `run_analytics` response. It will be stateless — it will have no access to the Analytics Engine, the SMR, or any execution backend. It will receive a self-contained DVL specification and return an image.

`vega2img` will not be required for consumers that can natively render DVL specifications. Agentic pipelines that produce static report output will be the primary use case.

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